

未命名 2

1 Kinematics

$$\begin{aligned}\bar{u}_\alpha(x_1, x_2) &= u_\alpha(x_1, x_2) - x_3 \phi_\alpha, & \bar{u}_3 &= w, \\ \epsilon_{\alpha\beta} &= \varepsilon_{\alpha\beta} + x_3 \kappa_{\alpha\beta}, \\ \varepsilon_{\alpha\beta} &= \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}), & \kappa_{\alpha\beta} &= -\frac{1}{2}(\phi_{\alpha,\beta} + \phi_{\beta,\alpha}), \\ \gamma_\alpha &= 2\epsilon_{\alpha 3} = w_{,\alpha} - \phi_\alpha.\end{aligned}$$

2 Constitutive relations

$$\begin{aligned}\sigma_{\alpha\beta} &= D_{\alpha\beta\gamma\eta} \epsilon_{\gamma\eta}, & \tau_{3\alpha} &= G \gamma_\alpha, \\ D_{\alpha\beta\gamma\eta} &= \frac{E}{1-\nu^2} \left(\nu \delta_{\alpha\beta} \delta_{\gamma\eta} + \frac{1-\nu}{2} (\delta_{\alpha\gamma} \delta_{\beta\eta} + \delta_{\alpha\eta} \delta_{\beta\gamma}) \right), \\ G &= \frac{E}{2(1+\nu)}.\end{aligned}$$

3 Stress resultants

$$\begin{aligned}N_{\alpha\beta} &= \int_{-h/2}^{h/2} \sigma_{\alpha\beta} dx_3 = h D_{\alpha\beta\gamma\eta} \epsilon_{\gamma\eta}, \\ M_{\alpha\beta} &= \int_{-h/2}^{h/2} x_3 \sigma_{\alpha\beta} dx_3 = \frac{h^3}{12} D_{\alpha\beta\gamma\eta} \kappa_{\gamma\eta}, \\ Q_\alpha &= k \int_{-h/2}^{h/2} \tau_{3\alpha} dx_3 = k G h \gamma_\alpha.\end{aligned}$$

4 Equilibrium equations and boundary conditions (strong form)

$$\delta\Pi = \int_{\Omega} \left(M_{\alpha\beta} \delta\kappa_{\alpha\beta} + Q_\alpha \gamma_\alpha - P_{\alpha\beta} w_{,\alpha} \delta w_{,\beta} \right) d\Omega = 0.$$

Integration by parts yields:

$$\begin{aligned}M_{\alpha\beta,\beta} - Q_\alpha &= 0 \quad \text{in } \Omega, \\ Q_{\alpha,\alpha} - (P_{\alpha\beta} w_{,\beta})_{,\alpha} &= 0 \quad \text{in } \Omega.\end{aligned}$$

Natural boundary conditions:

$$M_{\alpha\beta} n_\beta = \bar{M}_\alpha \quad \text{on } \Gamma_M, \quad (Q_\alpha - P_{\alpha\beta} w_{,\beta}) n_\alpha = \bar{Q} \quad \text{on } \Gamma_Q.$$

5 Galerkin weak form

Multiply each strong form equation by the corresponding test function δu_α , $\delta \phi_\alpha$, δw and integrate over Ω :

$$\int_{\Omega} (M_{\alpha\beta,\beta} - Q_\alpha) \delta \phi_\alpha d\Omega = 0, \quad \int_{\Omega} (Q_{\alpha,\alpha} - (P_{\alpha\beta} w_{,\beta})_{,\alpha}) \delta w d\Omega = 0.$$

Integrate by parts to transfer derivatives to the test functions:

$$\begin{aligned} \int_{\Omega} M_{\alpha\beta,\beta} \delta \phi_\alpha d\Omega - \int_{\Omega} Q_\alpha \delta \phi_\alpha d\Omega &= \int_{\Gamma} M_{\alpha\beta} n_\beta \delta \phi_\alpha d\Gamma - \int_{\Omega} M_{\alpha\beta} \delta \phi_{\alpha,\beta} d\Omega - \int_{\Omega} Q_\alpha \delta \phi_\alpha d\Omega = 0. \\ \int_{\Omega} Q_{\alpha,\alpha} \delta w d\Omega - \int_{\Omega} (P_{\alpha\beta} w_{,\beta})_{,\alpha} \delta w d\Omega &= \int_{\Gamma} Q_\alpha n_\alpha \delta w d\Gamma - \int_{\Omega} Q_\alpha \delta w_{,\alpha} d\Omega \\ &\quad - \int_{\Gamma} P_{\alpha\beta} w_{,\beta} n_\alpha \delta w d\Gamma + \int_{\Omega} P_{\alpha\beta} w_{,\beta} \delta w_{,\alpha} d\Omega = 0. \end{aligned}$$

Move all domain integrals to one side and boundary integrals to the other to obtain the weak form:

$$\int_{\Omega} (M_{\alpha\beta} \delta \phi_{\alpha,\beta} + Q_\alpha \delta \phi_\alpha + Q_\alpha \delta w_{,\alpha} - P_{\alpha\beta} w_{,\beta} \delta w_{,\alpha}) d\Omega = \int_{\Gamma_M} \bar{M}_\alpha \delta \phi_\alpha d\Gamma + \int_{\Gamma_Q} \bar{Q} \delta w d\Gamma.$$

Using the symmetry of strain and curvature:

$$M_{\alpha\beta} \delta \phi_{\alpha,\beta} = M_{\alpha\beta} \delta \kappa_{\alpha\beta}, \quad Q_\alpha \delta \phi_\alpha - Q_\alpha \delta w_{,\alpha} = Q_\alpha \delta \gamma_\alpha$$

$$\boxed{\int_{\Omega} (M_{\alpha\beta} \delta \kappa_{\alpha\beta} + Q_\alpha \delta \gamma_\alpha - P_{\alpha\beta} w_{,\alpha} \delta w_{,\beta}) d\Omega = 0.}$$

$$a_b + a_s - a_g = 0$$

6 Finite element discretization

$$w^h(x) = \sum_K N_K(x) w_K, \quad \phi_\alpha^h(x) = \sum_R N_R(x) \phi_{\alpha R}$$

$$\kappa_{\alpha\beta R} = -\frac{1}{2} (N_{R,\beta} \phi_{\alpha R} + N_{R,\alpha} \phi_{\beta R}), \quad \gamma_{\alpha K} = N_{K,\alpha} w_K - N_K \phi_{\alpha K}, \quad w_{,\alpha K} = N_{K,\alpha} w_K$$

$$a_b = \int_{\Omega} \frac{h^3}{12} D_{\alpha\beta\gamma\eta} \kappa_{\alpha\beta}^h \delta \kappa_{\gamma\eta}^h d\Omega$$

$$a_b = \sum_{R,S} \delta \phi_{iR} \left(\int_{\Omega} \frac{h^3}{48} D_{\alpha\beta\gamma\eta} (N_{R,\beta} \delta_{\alpha i} + N_{R,\alpha} \delta_{\beta i}) (N_{S,\eta} \delta_{\gamma j} + N_{S,\gamma} \delta_{\eta j}) d\Omega \right) \phi_{jS}$$

$$K_{RS}^b = \frac{h^3}{48} \int_{\Omega} D_{\alpha\beta\gamma\eta} (N_{R,\beta} \delta_{\alpha i} + N_{R,\alpha} \delta_{\beta i}) (N_{S,\eta} \delta_{\gamma j} + N_{S,\gamma} \delta_{\eta j}) d\Omega$$

$$a^b = \sum_{K,L} [\delta\phi_{1K} \quad \delta\phi_{2K}] \mathbf{K}_{KL}^b \begin{bmatrix} \phi_{1L} \\ \phi_{2L} \end{bmatrix}$$

$$a_s = \int_{\Omega} kGh \, \gamma_{\alpha}^h \, \delta\gamma_{\alpha}^h \, d\Omega$$

$$a_s = \sum_{K,L} \left[\delta w_K \left(\int_{\Omega} kGh \, N_{K,\alpha} N_{L,\alpha} \, d\Omega \right) w_L + \delta w_K \left(\int_{\Omega} -kGh \, N_{K,\alpha} N_L \, d\Omega \right) \phi_{\alpha L} \right. \\ \left. + \delta\phi_{\alpha K} \left(\int_{\Omega} -kGh \, N_K N_{L,\alpha} \, d\Omega \right) w_L + \delta\phi_{\alpha K} \left(\int_{\Omega} kGh \, N_K N_L \, \delta_{\alpha\beta} \, d\Omega \right) \phi_{\beta L} \right]$$

$$\mathbf{K}_{KL}^s = kGh \int_{\Omega} \begin{bmatrix} N_{K,\alpha} N_{L,\alpha} & -N_{K,\alpha} N_L \\ -N_K N_{L,\beta} & N_K N_L \delta_{\alpha\beta} \end{bmatrix} d\Omega$$

$$a_s = \sum_{K,L} [\delta w_K \quad \delta\phi_{\alpha K}] \mathbf{K}_{KL}^s \begin{bmatrix} w_L \\ \phi_{\beta L} \end{bmatrix}$$

$$a_g = P_{\alpha\beta} \int_{\Omega} w_{,\alpha} \delta w_{,\beta} \, d\Omega$$

$$a_g = \sum_{K,L} \delta w_K \left(P_{\alpha\beta} \int_{\Omega} N_{K,\alpha} N_{L,\beta} \, d\Omega \right) w_L$$

$$\mathbf{K}_{KL}^g = P_{\alpha\beta} \int_{\Omega} N_{K,\alpha} N_{L,\beta} \, d\Omega$$

$$\left(\begin{bmatrix} \mathbf{K}_{ww}^s & \mathbf{K}_{w\phi}^s \\ \mathbf{K}_{\phi w}^s & \mathbf{K}^b + \mathbf{K}_{\phi\phi}^s \end{bmatrix} - \begin{bmatrix} \mathbf{K}_{ww}^g & 0 \\ 0 & 0 \end{bmatrix} \right) \begin{bmatrix} \mathbf{d}_w \\ \mathbf{d}_\phi \end{bmatrix} = \mathbf{0}$$